

HAITAO XU

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EDUCATION

Cleveland State University, Cleveland, OH Fall 2021 - Summer 2025
Ph.D. in Computer Science GPA: 3.84/4.00
Dissertation: Algorithms for Facility Location Problems with Uncertainty on Graphs

Kunming University of Science and Technology, Kunming, China Fall 2011 - Spring 2014
M.S. in Software Engineering GPA: 85.7/100
Thesis: Research and Application of Software Component Technology in Examination System

Shandong Institute of Commerce and Technology, Jinan, China Fall 2005 - Spring 2008
A.S. in Computer Science and Applications GPA: 82.3/100

RESEARCH INTERESTS

Algorithms and Data Structures, Combinatorial Optimization, Operations Research

PUBLICATIONS

Peer Reviewed Journal Articles

1. **Haitao Xu** and Jingru Zhang. The k-Center Problem of Uncertain Points on Graphs. *Submitted to the Information Processing Letters*.
2. **Haitao Xu** and Jingru Zhang. The two-center problem of uncertain points on cactus graphs. *Journal of Combinatorial Optimization*, 49.4 (2025): 54.
3. **Haitao Xu** and Jingru Zhang. The two-center problem of uncertain points on a real line. *Journal of Combinatorial Optimization*, 45.2 (2023): 68.
4. Qi Sun, Liwen Jiang, **Haitao Xu**. A Double-Layer Combination Algorithm for Real-Time Information-Sharing Network Design Problem. *Complexity*, 2021(1), 4856593.
5. Qi Sun, Liwen Jiang, **Haitao Xu**. Expectation-Maximization Algorithm of Gaussian Mixture Model for Vehicle-Commodity Matching in Logistics Supply Chain. *Complexity*, 2021(1), 9305890.
6. Song, Yang, Yan-Qiu Liu, Qi Sun, Ming-Fei Chen, and **Hai-Tao Xu**. A Joint Optimization Model considering the Product User's Risk Preference for Supply System Disruption. *Mathematical Problems in Engineering*, no. 1 (2021): 5081753.

Peer Reviewed Conference Papers

1. **Haitao Xu** and Jingru Zhang. The two-center problem of uncertain points on trees. *In Proc. of the 16th Annual International Conference on Combinatorial Optimization and Applications (COCOAA)*, pages 485-697, 2023.
2. Lijie Wu, Qi Sun, **Haitao Xu**, Xiaochen Song, Yang Zhang. Design of hybrid simulated annealing algorithm for UAV scheduling based on coordinated task scheduling. *In Proc. of the 40th Chinese Control Conference (CCC)*, pages 1669-1674, 2021.

PROFESSIONAL EXPERIENCE

Cleveland State University, Cleveland, Ohio Fall 2021 - Present
Research Assistant: We study several facility location problems on various graphs related to uncertain demand data. Specifically, we focus on solving the two-center problem on a line, a tree graph, and a cactus

graph, as well as the k-center problem on a general graph. These center problems are classical combinatorial optimization problems and have been explored a lot in theoretical computer science. Our emphasis, however, lies in uncertain demand data, leading us to develop more general algorithmic approaches to address these problems.

Teaching Assistant: CIS 550 Introduction to Algorithms; CIS 606 Analysis of Algorithms; CIS 620 Advanced Operating Systems

KEDACOM, Shanghai, China

Apr 2016 - Feb 2020

C/C++ Software Engineer

Contributed to the development of core components for cross-platform video conferencing software, supporting Windows PCs, Android, and iOS devices. Component functionalities included data transmission, command distribution, and real-time messaging based on Google's Libjingle framework. Additionally served as a technical trainer for new team members, helping them onboard by introducing development workflows, domain knowledge, and the team's technical architecture.

Shanghai Huateng Software Systems Co Ltd, Shanghai, China

Aug 2014 - Apr 2016

C/C++ Software Engineer

Participated in the development of banking transaction systems, including batch trade settlement, debit card loss reporting, card replacement, and card security authentication. Also involved in deploying new modules to backend production servers in the banking environment.

Jinan Yuancheng Innovation Technology Co., LTD, Jinan, China

July 2008 - June 2010

Software Engineer

Participated in the development of an automated office system using C#, Ext JS, JavaScript, CSS, and Windows Workflow Foundation, primarily designed to support process review and approval workflows.

TECHNICAL SKILLS

1. Over 5 years of experience as a C/C++ developer, with a solid background in analysis, application development, system programming, and maintenance.
2. Experienced in software design and development using Python, C/C++, STL, multithreading, data structures, network programming, and design patterns.
3. Hands-on experience with Oracle database programming, including SQL, stored procedures, and related components.
4. Experience in web development using Java, JavaScript, HTML, and CSS.

SERVICES

Journal Reviewer

Hindawi Complexity

Conference Reviewer

The 40th Chinese Control Conference (CCC'2021)

Professional Activities

Being a Judge for Choose Ohio First Symposium at Cleveland State University, in Spring 2022.